

```
In [1]: # Import required libraries
import nltk
from nltk.tokenize import word_tokenize, TweetTokenizer, MWETokenizer
from nltk.tokenize.treebank import TreebankWordTokenizer
from nltk.stem import PorterStemmer, SnowballStemmer, WordNetLemmatizer
import string
```

```
In [2]: # Download required resources (run once)
```

```
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\rushi\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\rushi\AppData\Roaming\nltk_data...
[nltk_data] Downloading package omw-1.4 to
[nltk_data]   C:\Users\rushi\AppData\Roaming\nltk_data...
```

```
Out[2]: True
```

```
In [3]: # Sample text
```

```
text = "Hello! This is a sample sentence. Let's test tokenization, stemming, a

print("Original Text:\n", text)
print("="*60)
```

Original Text:

Hello! This is a sample sentence. Let's test tokenization, stemming, and lemmatization :) #NLP is fun!

=====

```
In [4]: # -----
# 1. Tokenization Techniques
# -----
```

```
# (a) Whitespace Tokenization
```

```
whitespace_tokens = text.split()
print("\nWhitespace Tokenization:\n", whitespace_tokens)
```

```
# (b) Punctuation-based Tokenization (remove punctuation)
```

```
translator = str.maketrans('', '', string.punctuation)
clean_text = text.translate(translator)
punct_tokens = clean_text.split()
print("\nPunctuation-based Tokenization:\n", punct_tokens)
```

```
# (c) Treebank Tokenization
```

```
treebank_tokenizer = TreebankWordTokenizer()
treebank_tokens = treebank_tokenizer.tokenize(text)
print("\nTreebank Tokenization:\n", treebank_tokens)
```

```
# (d) Tweet Tokenization
```

```
tweet_tokenizer = TweetTokenizer()
tweet_tokens = tweet_tokenizer.tokenize(text)
print("\nTweet Tokenization:\n", tweet_tokens)
```

```
# (e) MWE Tokenization (Multi-Word Expressions)
```

```
mwe_tokenizer = MWETokenizer([('New', 'York'), ('machine', 'learning')])
mwe_tokens = mwe_tokenizer.tokenize(word_tokenize(text))
print("\nMWE Tokenization:\n", mwe_tokens)
```

Whitespace Tokenization:

```
['Hello!', 'This', 'is', 'a', 'sample', 'sentence.', "Let's", 'test', 'tokeni-  
zation,', 'stemming,', 'and', 'lemmatization', ':)', '#NLP', 'is', 'fun!']
```

Punctuation-based Tokenization:

```
['Hello', 'This', 'is', 'a', 'sample', 'sentence', 'Lets', 'test', 'tokenizat-  
ion', 'stemming', 'and', 'lemmatization', 'NLP', 'is', 'fun']
```

Treebank Tokenization:

```
['Hello', '!', 'This', 'is', 'a', 'sample', 'sentence.', 'Let', "'s", 'test',  
'tokenization', ',', 'stemming', ',', 'and', 'lemmatization', ':', ')', '#',  
'NLP', 'is', 'fun', '!']
```

Tweet Tokenization:

```
['Hello', '!', 'This', 'is', 'a', 'sample', 'sentence', '.', "Let's", 'test',  
'tokenization', ',', 'stemming', ',', 'and', 'lemmatization', ':)', '#NLP', 'i-  
s', 'fun', '!']
```

MWE Tokenization:

```
['Hello', '!', 'This', 'is', 'a', 'sample', 'sentence', '.', 'Let', "'s", 'te-  
st', 'tokenization', ',', 'stemming', ',', 'and', 'lemmatization', ':', ')',  
'#', 'NLP', 'is', 'fun', '!']
```

```
In [5]: # -----  
# 2. Stemming  
# -----  
  
# Porter Stemmer  
porter = PorterStemmer()  
porter_stems = [porter.stem(word) for word in treebank_tokens]  
print("\nPorter Stemming:\n", porter_stems)  
  
# Snowball Stemmer  
snowball = SnowballStemmer("english")  
snowball_stems = [snowball.stem(word) for word in treebank_tokens]  
print("\nSnowball Stemming:\n", snowball_stems)
```

Porter Stemming:

```
['hello', '!', 'thi', 'is', 'a', 'sampl', 'sentence.', 'let', "'s", 'test',  
'token', ',', 'stem', ',', 'and', 'lemmat', ':', ')', '#', 'nlp', 'is', 'fun',  
'!']
```

Snowball Stemming:

```
['hello', '!', 'this', 'is', 'a', 'sampl', 'sentence.', 'let', "'s", 'test',  
'token', ',', 'stem', ',', 'and', 'lemmat', ':', ')', '#', 'nlp', 'is', 'fun',  
'!']
```

```
In [6]: # -----  
# 3. Lemmatization  
# -----  
  
lemmatizer = WordNetLemmatizer()  
lemmatized_words = [lemmatizer.lemmatize(word) for word in treebank_tokens]  
print("\nLemmatization:\n", lemmatized_words)
```

Lemmatization:

```
['Hello', '!', 'This', 'is', 'a', 'sample', 'sentence.', 'Let', "'s", 'test',  
'tokenization', ',', 'stemming', ',', 'and', 'lemmatization', ':', ')', '#',  
'NLP', 'is', 'fun', '!']
```

In []: